

MK-312-BT v1.4 — Test Results (fillable)

Black probe always = J1B (–) GND post. Click yellow boxes and type. Save the PDF when done.

Test 1 — Power rails (multimeter: 20 V DC)

#	What to probe (red lead)	Expected	Result	Pass/Fail
1.1	J1B (+) battery post	11.5 – 13.5 V		
1.2	JP4 — +12V test point	11.5 – 13.5 V		
1.3	JP3 — +5V test point	4.85 – 5.15 V		
1.4	AVR socket pin 11	0 V (GND)		
1.5	AVR socket pin 31	0 V (GND)		
1.6	AVR socket pin 10 (VCC)	4.85 – 5.15 V		
1.7	AVR socket pin 30 (AVCC)	4.85 – 5.15 V		
1.8	LCD pin 1 (VSS)	0 V		
1.9	LCD pin 2 (VDD)	4.85 – 5.15 V		
1.10	LCD pin 3 (V0) — sweep VR4	0 V → 5 V across sweep		

Test 2 — AVR socket → LCD continuity (POWER OFF, AVR REMOVED, ribbon CONNECTED)

#	What to probe (red lead)	Expected	Result	Pass/Fail
2.1	AVR pin 23 (PC1) → LCD pin 5 (R/W)	BEEP		
2.2	AVR pin 24 (PC2) → LCD pin 6 (E)	BEEP		
2.3	AVR pin 25 (PC3) → LCD pin 4 (RS)	BEEP		
2.4	AVR pin 26 (PC4) → LCD pin 11 (DB4)	BEEP		
2.5	AVR pin 27 (PC5) → LCD pin 12 (DB5)	BEEP		
2.6	AVR pin 28 (PC6) → LCD pin 13 (DB6)	BEEP		
2.7	AVR pin 29 (PC7) → LCD pin 14 (DB7)	BEEP		

If any 2.x fails to BEEP, that signal's path is broken — check J11 and J10 pins, then trace.

Test 3 — Solder bridge between adjacent pins (POWER OFF, AVR REMOVED)

#	What to probe (red lead)	Expected	Result	Pass/Fail
3.1	AVR pin 22 ↔ 23 (PC0 / PC1)	NO beep		
3.2	AVR pin 23 ↔ 24 (PC1 / PC2)	NO beep		
3.3	AVR pin 24 ↔ 25 (PC2 / PC3)	NO beep		
3.4	AVR pin 25 ↔ 26 (PC3 / PC4)	NO beep		
3.5	AVR pin 26 ↔ 27 (PC4 / PC5)	NO beep		
3.6	AVR pin 27 ↔ 28 (PC5 / PC6)	NO beep		
3.7	AVR pin 28 ↔ 29 (PC6 / PC7)	NO beep		

Test 4 — LCD power, contrast, backlight (POWER ON)

#	What to probe (red lead)	Expected	Result	Pass/Fail
4.1	Backlight glow (look in dim light)	YES		
4.2	Sweep VR4 fully both directions	Blocks fade dark↔light		
4.3	LCD pin 3 (V0) voltage as VR4 sweeps	~0 V → ~5 V		
4.4	LCD DB4–7 floating (AVR out, probe 11–14)	≈ 0 V each		

Notes / observations